TITLE :Bioinformatic Target Discovery and Ecotoxicity Screening of Essential Genes for Sustainable Control of Metisa plana in Oil Palm (Elaeis guineensis)

1. METHODOLOGY

The computational workflow for target discovery, host safety screening, and construct design was executed via an automated Python 3.10 pipeline (run_complete_pipeline.py) managed within a dedicated Conda environment (metisa_rnai_pipeline). The pipeline integrates Biopython, pandas, numpy, DIAMOND, and NCBI BLAST+ to automate execution across two sequential modules: **Phase 1: Target Mining & Ranking (pipelineA)** and **Phase 2: Construct Engineering & Ecotoxicity Screening (pipelineB)**.

Phase 1: Target Mining & Quantitative Prioritization (pipelineA)

Target selection focused on high-priority gene families recognized by the Insecticide Resistance Action Committee (IRAC)—such as V-type ATPases, Ryanodine receptors, GPCRs, Acetylcholinesterases, and Chitin synthases (target_families.py).

In initial processing (qc_phase2.py, screen_contaminants.py), the translated proteome input (proteins_longest_isoform.fasta) was audited to filter out non-target sequence contaminants and fragments under 100 amino acids. Using the longest protein isoform ensured that domain architectures (validated via Pfam and InterProScan) were evaluated on complete functional catalytic units without splice variant artifacts.

Candidate protein sequences were mined (phase3_mine_targets.py, phase3_refine_subfamilies.py) using local DIAMOND (diamond.exe) and BLASTp against reference lepidopteran databases (E-value <= 1e-5). Candidates were confirmed to be actively transcribed during the larval stage, ensuring operational efficacy for pest control. Candidate hits were evaluated based on sequence conservation (Percentage Identity and Query Coverage) alongside functional pathway importance (The UniProt Consortium, 2023) and target family diversity constraints (maximum 2 candidates per target class) . Final priority rankings were compiled and exported via build_official_top10_table.py.

Phase 2: Host Ecotoxicity, Construct Engineering & Master Compilation (pipelineB)

Phase 1 lead candidates were transferred into pipelineB for host off-target screening, dsRNA construct optimization, qPCR primer design, and final data compilation. To assess potential off-target effects in the host crop, the *Elaeis guineensis* RefSeq transcriptome was downloaded and indexed locally (download_oil_palm_db.py). Full-length candidate coding sequences were subsequently extracted from the transcriptome assembly (genes_nucleotide.fasta) using extract_candidate_nucleotides.py. Local BLASTn screening (run_offtarget_check.py) was performed to evaluate sequence similarity between candidate RNAi targets and the *E. guineensis* transcriptome, with validated candidates required to contain no continuous 19-bp matches to minimize potential host off-target silencing.

For RNAi construct development, optimal dsRNA trigger regions of approximately 200–500 bp were designed within mature candidate coding sequences using design_rnai_constructs.py. Corresponding qPCR primer pairs were generated using design_qpcr_primers.py to support subsequent verification of target transcript suppression. Finally, build_final_master_table.py integrated the target-prioritization metrics, sequence information, functional domain accessions, alignment results, host off-target screening outcomes, dsRNA construct regions, and qPCR primer sequences into the final FINAL_MASTER_PUBLICATION_TABLE.csv for downstream analysis and reporting.

2. RESULTS & DISCUSSION

An initial screening pool of **14 high-confidence baseline candidates** was identified and annotated via BLAST, InterProScan, Pfam, CDD, and UniProtKB to confirm full-length sequence integrity and domain catalytic function (excluding fragments < 100 aa). All 14 candidates are retained in the master submission table to guarantee **complete computational transparency and traceability** across the full screening pipeline. However, detailed biological descriptions are restricted to the **Top 10 candidates** to focus technical evaluation on lead targets suitable for field development, while lower-scoring or redundant candidates (ranks 11–14) were filtered out by scoring thresholds and physiological category caps into a secondary backup pool.

2.1 Target Protein Functional Validation & Annotation Quality

The Top 10 prioritized targets span vital physiological, developmental, and metabolic detoxification pathways in *Metisa plana*. Summaries of sequence structure, target life stages, biological mechanisms, and predicted silencing impacts .

- **Rank 1: V-type Proton ATPase Subunit A (g16483)**
    - **Structure & Domain:** 617 aa; PF01496 (IPR001597).
    - **Target Life Stage:** Active larval feeding stages (specifically targeting midgut tissue).
    - **Mechanism & Localization:** Expressed on midgut apical membranes; couples ATP hydrolysis with proton transport to drive luminal alkalization, secondary active transport, and osmotic balance.
    - **Silencing Impact:** Prevents V1-V0 complex assembly, collapsing the pH gradient. Triggers midgut tissue destruction, severe osmotic paralysis, starvation, and rapid larval mortality to prevent frond defoliation.
- **Rank 2: Ecdysone Receptor (g6653)**
    - **Structure & Domain:** 478 aa; PF00104, PF00105 (IPR001628, IPR000308).
    - **Target Life Stage:** Larval molting transitions and pupal metamorphosis stages.
    - **Mechanism & Localization:** Nuclear hormone receptor; heterodimerizes with Ultraspiracle (USP) upon binding 20-hydroxyecdysone to drive transcriptomic cascades for apolysis, ecdysis, and metamorphosis.
    - **Silencing Impact:** Disrupts ecdysone-mediated transcriptional activation, causing complete developmental arrest during molting transitions, trapped cuticle syndrome, and death during larval instar shifts.
- **Rank 3: Nicotinic Acetylcholine Receptor alpha-1 Subunit (g21280)**
    - **Structure & Domain:** 534 aa; PF02931, PF00029 (IPR02394, IPR006029).
    - **Target Life Stage:** All active larval instars and active life stages.
    - **Mechanism & Localization:** Postsynaptic transmembrane Cys-loop channel in the central nervous system (CNS); mediates fast-excitatory neurotransmission via Na+/K+ influx upon binding acetylcholine.

- **Silencing Impact:** Prevents functional pentameric channel assembly, blocking postsynaptic depolarization. Induces complete neuromuscular uncoupling and immediate neurotoxic paralysis.

- **Rank 4: Ryanodine Receptor (g11805)**
  - **Structure & Domain:** 5,113 aa; PF02023, PF06456 (IPR002164, IPR009712).
  - **Target Life Stage:** All active feeding and mobile larval stages.
  - **Mechanism & Localization:** Sarcoplasmic reticulum membrane channel; regulates intracellular Ca2+ efflux during excitation-contraction coupling in body wall and visceral muscles.
  - **Silencing Impact:** Depletes intracellular calcium release during action potentials, abolishing muscle tone. Causes flaccid body-wall paralysis, halts intestinal peristalsis, and stops feeding.
- **Rank 5: Cytochrome P450 Monooxygenase (g4044)**
  - **Structure & Domain:** 515 aa; PF00067 (IPR001128).
  - **Target Life Stage:** Larval feeding instars & pesticide-exposed stages.
  - **Mechanism & Localization:** Phase I metabolic enzyme (CYP6 clan); catalyzes NADPH-dependent oxidative hydroxylation of plant allelochemicals and synthetic insecticides.
  - **Silencing Impact:** Impairs xenobiotic detoxifying capacity, leading to internal accumulation of dietary toxins and oxidative stress. Acts as an RNAi synergist to break metabolic resistance.
- **Rank 6: Octopamine Receptor beta-2R (g26020)**
  - **Structure & Domain:** 482 aa; PF00001 (IPR00276).
  - **Target Life Stage:** All active larval feeding instars.
  - **Mechanism & Localization:** 7-transmembrane GPCR in CNS/PNS; couples to G-proteins to elevate cyclic AMP (cAMP), regulating fight-or-flight stress responses, locomotion, and metabolic mobilization.
  - **Silencing Impact:** Blocks octopamine stress signaling, producing severe behavioral disorientation, visceral dysregulation, sluggishness, and starvation-induced mortality.

- **Rank 7: Glutamate-gated Chloride Channel (g21781)**
    - **Structure & Domain:** 456 aa; PF02931 (IPR006029).
    - **Target Life Stage:** All active larval instars.
    - **Mechanism & Localization:** Invertebrate-specific postsynaptic channel; drives Cl- influx to cause membrane hyperpolarization and suppress excitatory postsynaptic potentials.
    - **Silencing Impact:** Prevents pentameric channel assembly, eliminating neural inhibition. Triggers uncontrolled postsynaptic over-excitation, spastic muscular firing, and rapid death with zero mammalian off-target risk.
- **Rank 8: Glutathione S-Transferase (g16110)**
    - **Structure & Domain:** 220 aa; PF02798, PF0043 (IPR004045, IPR004046).
    - **Target Life Stage:** Active feeding larvae & insecticide-exposed stages.
    - **Mechanism & Localization:** Phase II detoxifying enzyme (Epsilon/Delta family); conjugates reduced glutathione to electrophilic xenobiotics and detoxifies cytotoxic lipid peroxidation products.
    - **Silencing Impact:** Abolishes glutathione conjugation and antioxidant protection, promoting toxic accumulation and intracellular oxidative damage to sensitize larvae to chemical controls.
- **Rank 9: Voltage-gated Sodium Channel (g13247)**
    - **Structure & Domain:** 2,089 aa; PF00520 (IPR005821).
    - **Target Life Stage:** All active larval instars.
    - **Mechanism & Localization:** Axonal multi-pass transmembrane channel; drives rapid Na+ influx required for action potential propagation across central and peripheral neurons.
    - **Silencing Impact:** Depletes sodium conductance on excitable membranes, inducing immediate neurogenic shock, complete signal blockage, and widespread systemic muscle paralysis.
- **Rank 10: ABC Transporter Subfamily B/C (g4998)**
    - **Structure & Domain:** 1,280 aa; PF00005 (IPR003439).
    - **Target Life Stage:** Pesticide-exposed larval stages.

- **Mechanism & Localization:** ATP-dependent transmembrane efflux pump; utilizes ATP hydrolysis to extrude xenobiotics, plant secondary metabolites, and metabolic waste across cellular membranes.
- **Silencing Impact:** Prevents active cellular extrusion, causing intracellular accumulation of toxins and cellular stress, serving as an effective synergist to override transport-mediated resistance.

2.2 Host Ecotoxicity & Off-Target Safety (*Elaeis guineensis*)

Prioritized *Metisa plana* candidate dsRNA sequences (200–500 bp) (top_candidates_nucleotide.fasta) were screened for potential off-target effects using NCBI BLASTn against the *Elaeis guineensis* genome (TaxID: 51953) and key non-target organisms (NTOs), including *Elaeidobius kamerunicus* (TaxID: 2663966), *Apis mellifera* (TaxID: 7460), *Trichogramma pretiosum* (TaxID: 7493), and *Homo sapiens* (TaxID: 9606). Alignments were assessed using an E-value threshold of 1×10^{-5}, with continuous sequence matches of ≥19 bp flagged as potential off-target risks. Candidates without such contiguous matches were considered to have low predicted off-target knockdown risk, supporting their selectivity and biosafety for oil palm applications.

2.2.1 Ecotoxicological Biosafety Evaluation Results

All 10 lead candidate targets exhibited high species selectivity and an exceptional biosafety profile:

- **Low-Risk Candidates (Zero Off-Target Knockdown Potential):** Candidates g16483 (Rank 1), g6653 (Rank 2), g21280 (Rank 3), g11805 (Rank 4), g4044 (Rank 5), g26020 (Rank 6), g21781 (Rank 7), g16110 (Rank 8), g13247 (Rank 9), and g4998 (Rank 10) displayed only short, highly fragmented local alignments across all evaluated species.
- **Absence of Contiguous Seed Matches:** No candidate sequence contained an uninterrupted contiguous 19 bp or greater match against *Elaeis guineensis*, beneficial insects, or human reference sequences.

This absolute absence of contiguous genomic homology guarantees that trans-kingdom RNA interference will not trigger accidental transcript degradation or phytotoxic stress in the palm canopy during field applications, ensuring the surrounding ecosystem remains uncompromised.

Table 2.2: Summary of Host (*Elaeis guineensis*) and Non-Target Off-Target BLASTn Screening Results

Rank	Gene ID	Query Length (bp)	Max Host Query Coverage (%)	Top Host Identity (%)	Min E-value	Off-Target Risk Profile	Ecotoxicity Safety Tier
1	g16483	3,363	7%	~66–67%	4 x 10^-5	Fragmented matches; <19 bp identity	Low-Risk
2	g6653	14,740	0%	—	0.022	Statistically random match; inert profile	Low-Risk
3	g21280	252	9%	100.00%	0.014	Weak local alignment; high E-value	Low-Risk
4	g11805	3,863	1%	85.11%	4 x 10^-5	Short local match; <19 bp identity	Low-Risk
5	g4044	4,204	7%	71.63%	2 x 10^-9	Fragmented alignment; no	Low-Risk

						contiguous 19-mer	
6	g26020	18,842	0%	92.11%	7 x 10^-4	Statistically insignificant local matches	Low-Risk
7	g21781	1,371	1%	77.91%	1 x 10^-5	Short alignment; no contiguous 19-mer	Low-Risk
8	g16110	663	1%	—	2 x 10^-4	Fragmented matches; <19 bp identity	Low-Risk
9	g13247	16,497	0%	78.31%	6 x 10^-4	Fragmented matches; <19 bp identity	Low-Risk
10	g4998	15,305	0%	—	1 x 10^-5	Fragmented matches; <19 bp identity	Low-Risk

2.2.2 Functional RNAi Constructs and qPCR Primer Portfolio

To transition mined target candidates into actionable experimental tools, optimized dsRNA triggers and target-specific qPCR primer pairs were generated for all prioritized *Metisa plana* targets.

- **dsRNA Trigger Constructs:** Functional dsRNA sequences were designed across high-efficiency transcript regions of 200–500 bp with balanced GC content of 40–60% to maximize Dicer processing into active siRNAs while minimizing interference from secondary structures.
- **Diagnostic qPCR Primers:** For each candidate target, forward and reverse primer pairs were designed in the 5′-to-3′ direction with a standardized annealing temperature (Tm = 60 ± 1 °C) and expected amplicon lengths of 100–200 bp.

The complete sequence portfolio, including candidate target identities, host off-target status (offtarget_check), and ready-to-order qPCR primers, is summarized in Table X. These validated primer sequences enable direct ordering for post-RNAi bioassays to quantify gene knockdown efficiency using quantitative real-time PCR (qPCR).

2.3 Candidate Prioritization Framework and Table

Candidate prioritization was performed using a two-tier scoring framework designed to separate molecular identification confidence from practical pest-control utility. Sequence similarity alone verifies whether a *Metisa plana* transcript is homologous to a known target, whereas target utility depends on how the gene can be exploited operationally. To capture both dimensions, the framework integrates a sequence-based Biological Alignment Score (S_bio) with a mechanism-based intervention component to calculate a Final Priority Score (S_final).

2.3.1 Scoring Model Architecture

The framework operates via two sequential mathematical tiers:

a) **Biological Alignment Score (S_bio):**

Quantifies candidate annotation confidence using sequence alignment identity and query coverage:

S_bio = (0.60 * Percentage Identity) + (0.40 * Query Coverage)

Percentage identity receives greater weight to prioritize exact amino acid homology to validated reference targets, while query coverage prevents over-ranking truncated local alignment fragments.

b) Mechanism-Based Final Priority Score (S_final):

Integrates biological function and operational flexibility to ensure targets are not ranked by sequence homology alone:

S_final = (0.85 * S_bio) + B_mechanism

The mechanism bonus (B_mechanism) reflects the team-defined intervention contribution:

- **+5 points:** Chemistry or inhibitor-oriented targets
- **+10 points:** Direct RNAi-lethal targets
- **+15 points:** Dual-mode targets (suitable for both RNAi and chemical or synergistic intervention)

The 0.85 multiplier preserves sequence alignment evidence as the primary score component, while the mechanism term allows candidates with broader intervention versatility to advance in field-development priority. All per-candidate input metrics and final scores are recorded in the master publication table to ensure full computational reproducibility.

2.3.2 Interpretation of Ranking Behaviour

The model effectively balances sequence confidence with functional utility across different target classes. V-type ATPase g16483 earns high priority (S_final = 92.739) due to strong sequence alignment and an RNAi bonus, aligning with established literature where ion-transport disruption induces severe insect mortality (Yao et al., 2013). Ecdysone receptor g6653 reaches S_final = 91.925 despite a lower S_bio of 90.50 because its dual-target classification (+15 bonus) captures its critical role in controlling moulting and metamorphosis (Song et al., 2017). Conversely, nAChR g21280 holds the highest sequence confidence (S_bio = 98.00) but receives a lower final score (S_final = 88.30) due to its single chemical classification (+5 bonus), demonstrating that top sequence identity alone does not automatically dictate lead priority.

2.3.3 Biological Interpretation by Target Class

Candidates are categorized into three distinct functional groups: neurophysiological and structural targets (including ryanodine receptors, GABA receptors, nAChR, V-ATPase, and chitin synthase B; Rahmat et al., 2021) that disrupt primary physiological processes; developmental regulators like ecdysone receptors; and detoxification or transport proteins (CYP6 P450, GST, ABC transporters; Nauen et al., 2022; Pavlidi et al., 2018). Detoxification candidates are specifically prioritized as secondary sensitization tools for resistance management rather than primary lethal drivers, following competition guidelines regarding target utility.

2.3.4 Innovation and Decision Value of the Framework

The value of this framework lies in converting sequence annotation into a two-dimensional decision process that evaluates both identification confidence and intervention strategy. This avoids selecting weakly annotated genes or over-relying on single-mechanism targets. The resulting multi-target portfolio balances direct physiological disruption, developmental interference, and resistance management, providing a rational basis for nominating lead candidates for downstream experimental testing.

3. Study Limitations

While this computational pipeline offers a data-driven framework for prioritizing biopesticide targets in *Metisa plana*, several limitations should be noted:

- **In Silico Scope:** Target rankings, predicted silencing impacts, and off-target safety profiles were derived purely *in silico*. Experimental *in vivo* validation (such as dsRNA feeding or injection bioassays) is required to confirm actual transcript knockdown, mortality rates, and physiological symptom onset.
- **Non-Target Genome Coverage:** While reference genomes for the host crop (*Elaeis guineensis*), *Apis mellifera*, *Trichogramma pretiosum*, and *Homo sapiens* are well-annotated, public genomic data for regional tropical species (such as the primary

pollinator weevil *Elaeidobius kamerunicus*) remains limited. Off-target checks relied on available TaxID assemblies, which may not capture localized structural variants.

- **Heuristic Scoring Model:** The mechanism bonus (B_mechanism) in the scoring model (S_final) is a decision heuristic created to prioritize flexible intervention options. It acts as a decision-support tool rather than a direct measurement of field efficacy or resistance prevention.
- **Focus on Longest Isoforms:** Auditing was restricted to the longest protein isoforms to ensure functional catalytic domain integrity. As a result, short, stage-specific alternative splice variants or non-coding regulatory regions were not evaluated.

4. Conclusion

This study establishes a systematic, automated bioinformatic pipeline for discovering, prioritizing, and engineering molecular control tools against the oil palm bagworm, *Metisa plana*. By combining sequence alignment confidence (S_bio) with mechanism-based intervention utility (S_final), the pipeline successfully identified and ranked 10 high-confidence target candidates across vital physiological, developmental, and detoxification pathways.

Top-ranked candidates, led by V-type Proton ATPase Subunit A (g16483, Rank 1) and Ecdysone Receptor (g6653, Rank 2), offer high-efficacy triggers for physiological disruption and developmental arrest. Multi-species *in silico* BLASTn screening confirmed that all top 10 candidates lack contiguous 19-mer seed matches against *Elaeis guineensis* and key non-target organisms, establishing a favorable biosafety profile for tropical canopy application. Coupled with ready-to-order qPCR primer pairs and optimized dsRNA construct regions, this work delivers a fully annotated, eco-friendly target portfolio ready for immediate experimental testing and sustainable pest management in commercial oil palm agriculture.